Clear Creek TRNG

Colorado School of Mines

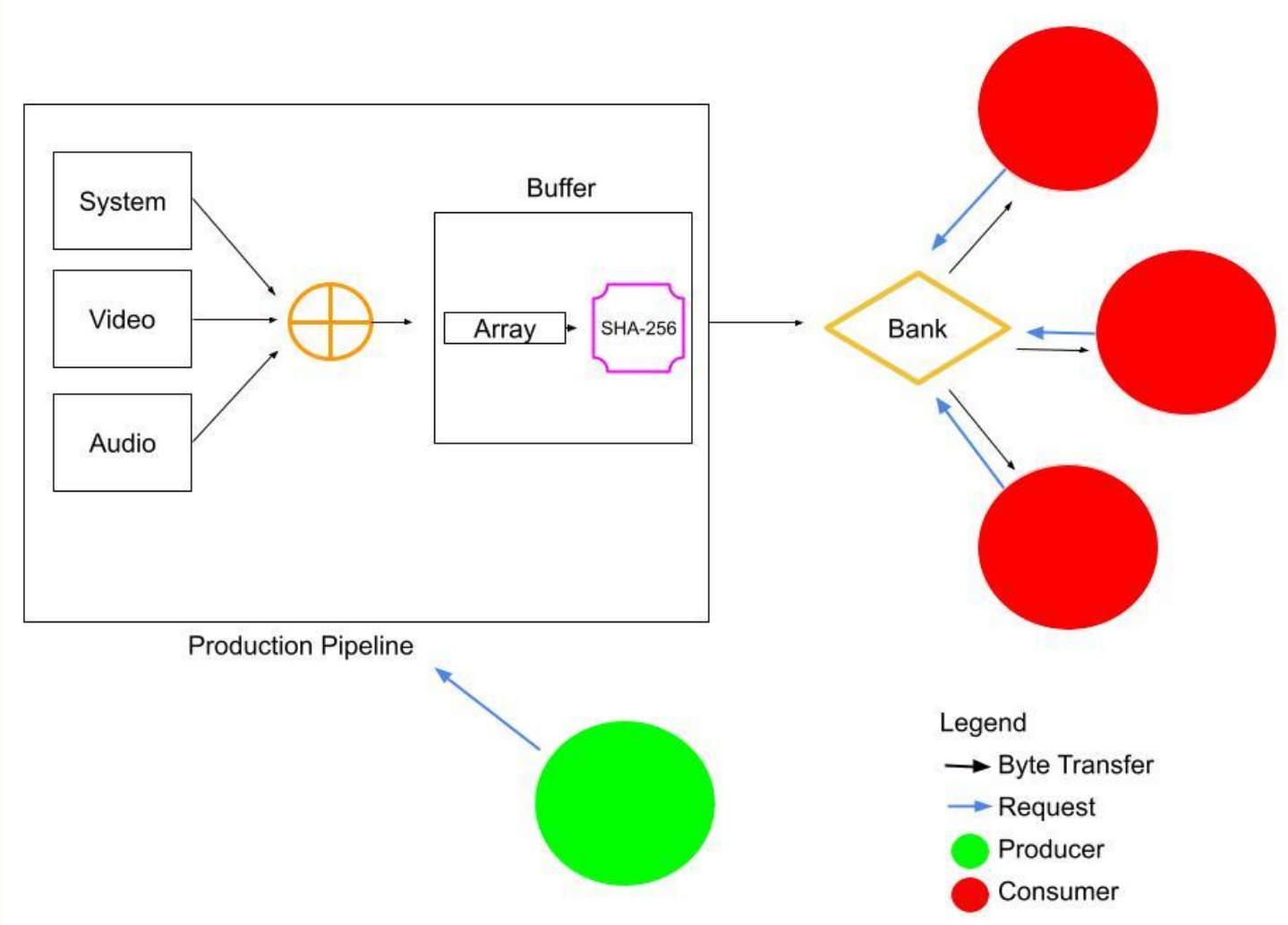
Introduction + Motivation

System Requirements

- Our output must pass the NIST Tests for Secure Random Number Generation
  - NIST is a trusted authority that sets rigorous standards for cryptographic security
  - Passing NIST tests provides a clear validation of the quality of our random output
- Entropy from multiple sources must be properly buffered and mixed

  Random outputs must be cryptographically mixed (e.g., via SHA-256 hashing) before storage
- The system must ensure sufficient entropy accumulation before random bytes are finalized
- Generated random bytes must be securely stored and managed
- Strict access control to the system must be enforced
- Communication must defend against man-in-the-middle (MITM) attacks during byte transfers

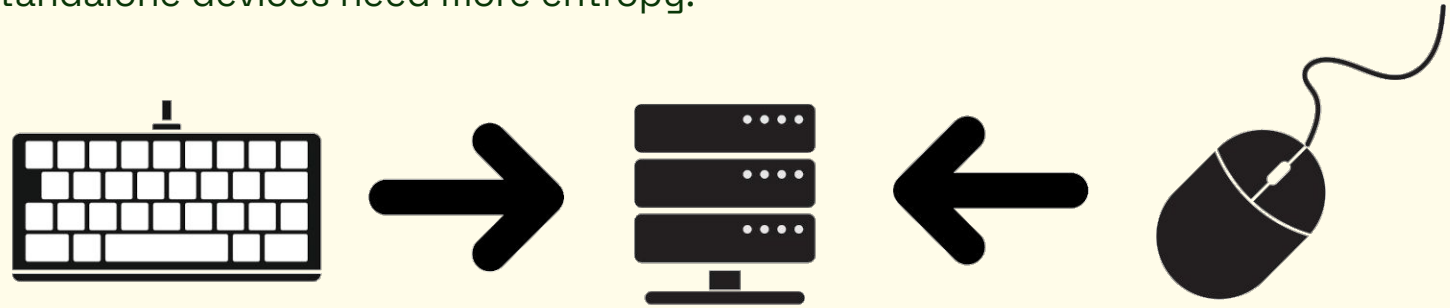
System Overview



System Entropy

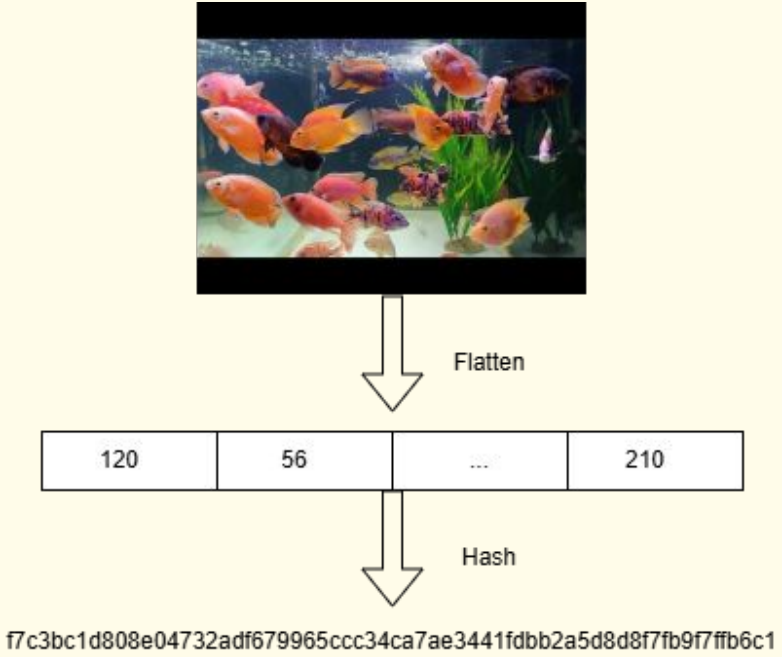
- Commonly gathered from system environmental noise such as:
  - Key press timings
  - Mouse movements
  - Interrupt timings
- Sufficient entropy for home computers with multiple peripherals
- Insufficient for standalone systems and servers

These standalone devices need more entropy!



Video Entropy

- High Variability in Pixel Data
  - Even slight movements like a fish swimming by, a ripple in water, or changes in lighting can cause unpredictable shifts in pixel data. This results in a large and constantly changing entropy space.
- Real World Scenes
  - Introduces entropy from real world physical phenomena like biological movement, reflections, motion, etc.
- Noise and Compression
  - Adds additional entropy

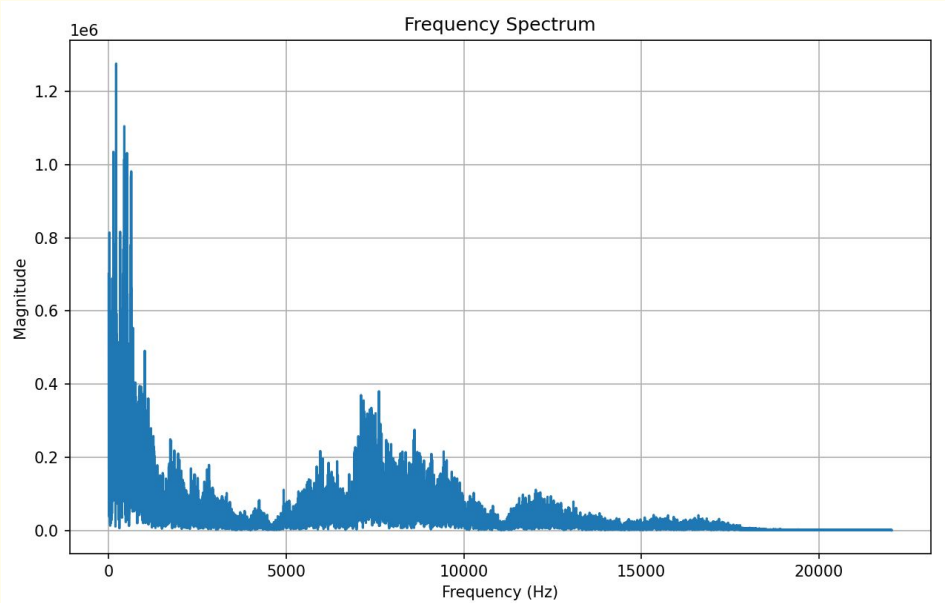


Sound Entropy

Sound/Recording Devices Give **Two** Sources of Entropy

1) **Environmental** Noise
   a) The ambient sound around the microphone
   b) **Drawback:** Quiet Spaces
2) Analog to Digital Signal **Conversion**
   a) **Thermal** and **Quantization** noise
   b) Increasing the **gain** of a microphone <u>increases</u> the noise by <u>decreasing</u> stability

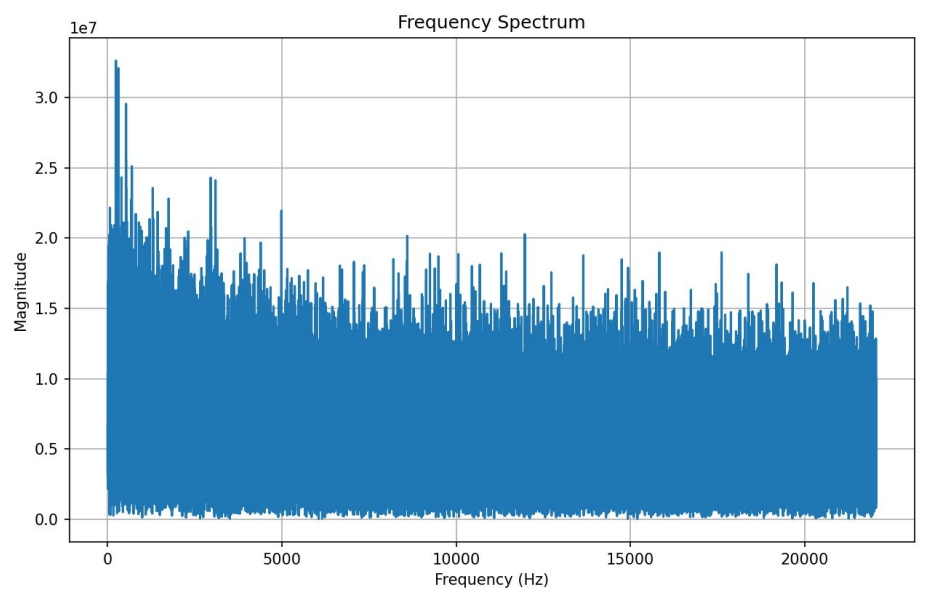
* We want the distribution of frequencies to be as random as reasonably possible



Sound Entropy

Strategies for **Maximizing** Entropy:

1) Random **Audio** (environment dependent)
2) Random **Noise** (microphone dependent)
3) **Amplify** the Signal
   a) Adds **Chaos**
4) High **Sampling Rate**
   a) Decreases necessary **recording duration**, improving **response time**
   b) Captures **subtle** randomness
5) Microphone **Frequency Response**
   a) Larger **range** captures more data
6) Compression that benefits from the **Avalanche Effect** (Sha3-256 for us)

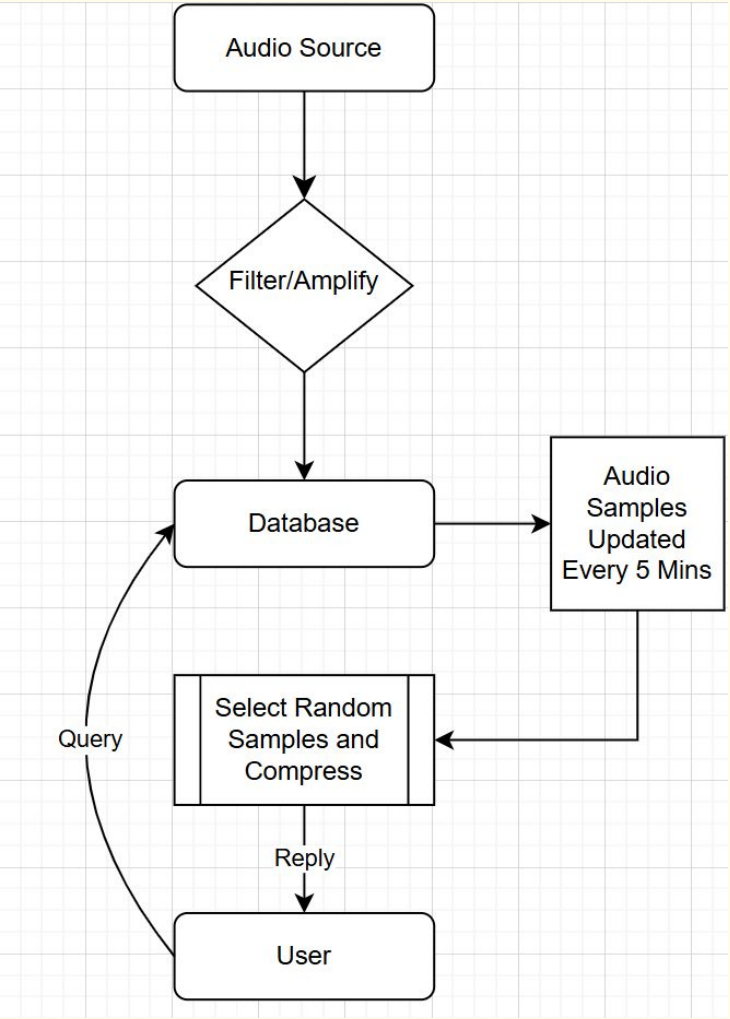


Sound Entropy –Improvements

Don't rely on actively recording audio

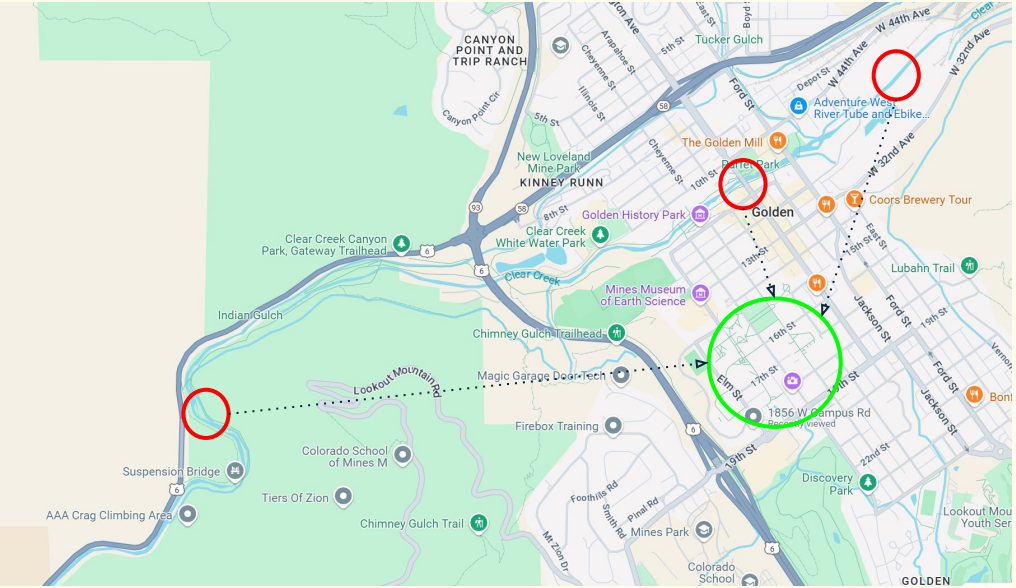
Store a **database** of audio samples that are chosen at random instead

1) Record continuously and update the database every time quantum
2) Querying for a key can be **near instant**
   a) Use a **random seed** for selecting recording/frequency



Future Plans + Limitations

- Add more Raspberry Pis along Clear Creek
  - Coors Factory for example
- Connect Pis to a central server to handle TRN distribution
  - Allow Pis to securely transmit byte streams to server
  - Enable requesting of TRN from server



Thank You

References

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